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Mammalian organ regeneration in spiny mice

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Abstract

Fibrosis-driven solid organ failure is a major world-wide health burden with few therapeutic options. Spiny mice (genus: *Acomys*) are terrestrial mammals that regenerate severe skin wounds without fibrotic scars to evade predators. Recent studies have shown that spiny mice also regenerate acute ischemic and traumatic injuries to kidney, heart, spinal cord, and skeletal muscle. A common feature of this evolved wound healing response is a lack of formation of fibrotic scar tissue that degrades organ function, inhibits regeneration, and leads to organ failure. Complex tissue regeneration is an extremely rare property among mammalian species. In this article, we discuss the evidence that *Acomys* represents an emerging model organism that offers a unique opportunity for the biomedical community to investigate and clinically translate molecular mechanisms of scarless wound healing and regeneration of organ function in a mammalian species.

Keywords

Acomys ; Wound healing; Fibrosis; Evolution

Introduction

Many unique phenotypes in nature represent evolved adaptations to become either a better predator, a more elusive prey, or a more attractive mate (Endler 1986; Sterling and McCreless 2006; Conner and Corcoran 2012; Pembury Smith and Ruxton 2020). African

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spiny mice (genus: *Acomys*) are muroid rodent species native to arid and rocky habitats in Africa and the Middle East that evolved a number of unique adaptations to be more elusive prey animals (Fig. 1). Their major predators are foxes, snakes, and raptors (Jones and Dayan 2000; Regula 2012). A deterrent against soft-mouthed predators is provided by the characteristic stiff spiny hairs that cover the dorsum of *Acomys* species, thus the common name “spiny mice” (Montandon et al. 2014; Jiang et al. 2019). Predatory raptors, however, are less likely to be deterred in this manner necessitating a different strategy to avoid predation. In this case, the property of skin shedding provides spiny mice with an effective avoidance capability (Seifert et al. 2012). The acquisition of this trait seems to have occurred along with a suite of phenotypes that permit long-term success of this strategy for species survival. These phenotypes include absence of infection in exposed subdermal tissues, lack of pain perception upon skin shedding, minimal bleeding, and failure to form scar tissue (Seifert et al. 2012; Maden and Brant 2019; Brewer et al. 2021). Scarless healing is accompanied by complete regeneration of skin including de novo hair follicles, sebaceous glands, cartilage, adipose tissue, nerves, and blood vessels in the correct architecture for restoration of structure and function of skin tissue (Seifert et al. 2012; Brant et al. 2015; Gawriluk et al. 2016; Matias Santos et al. 2016; Jiang et al. 2019; Maden and Brant 2019; Harn et al. 2021; Brewer et al., 2021).

It is this latter adaptation that has captured the attention of scientists worldwide (Maden and Varholick 2020; Gaire et al. 2021; Koopmans et al. 2021; Nogueira-Rodrigues et al. 2022; Okamura et al. 2022). Regeneration of complex tissue is extremely rare in adult mammals (Poss 2010; Maden 2018). Yet, most mammalian species, including humans, exhibit scarless regeneration in fetal or neonatal life (Colwell et al. 2003; Drenckhahn et al. 2008; Pratsinis et al. 2019; Porrello et al. 2011; Abrams et al. 2021). This may speak to the possibility that regeneration is a more broadly distributed property of mammalian genomes than is generally thought based on adult phenotypes of the comparatively few mammalian species in which it has been examined (Bely and Nyberg 2010; Andres-Mateos et al. 2012; Abrams et al. 2021; Murugan et al. 2022). If so, then the question becomes what are the mechanisms that repress this property in non-regenerating mammals and how can this potential be brought forward from its cryptic and repressed state (Bely and Sikes 2010; Cui et al. 2020; Williams et al. 2021; Abrams et al. 2021)? The fact that regeneration is both ancient (Fröbisch et al. 2014) and current, as the property in adults is now found scattered throughout the animal kingdom (Brookes et al. 2001), highlights its resiliency and suggests that strong selection pressures may be required to maintain expression in adult life. The cases of human fingertip regeneration (Illingworth 1974; Allan et al. 2006), as well as regeneration of surgical wounds in human fetal life (Colwell et al. 2003), suggest that this potential also resides in the human genome. Taken together, these findings argue that the information for organ regeneration may broadly exist in mammalian genomes but the expression of this information is efficiently silenced around the time of birth. Therefore, the possibility that spiny mice have avoided silencing this latent potential suggests that further study of this animal’s remarkable wound healing properties may reveal clues as to the silencing mechanism and its possible reversal in non-regenerating mammalian species.

Spiny mouse as an emerging model organism

Collectively, most investigators use only a small number of metazoan model organisms for their research (Goldstein and King 2016). These familiar models include mice, rats, zebrafish, *Xenopus*, *C. elegans*, and *Drosophila*. As an emerging model organism, spiny mice exhibit several unique phenotypes that are not found in traditional mammalian model organisms used for research (Gaire et al. 2021). In addition to the scarless wound healing trait introduced above, Bellofiore et al. reported that spiny mice are the first rodent species identified to exhibit menstruation (Bellofiore et al. 2017, 2021) thus offering a new genetic landscape for interrogation of the physiology and reproductive biology of mammalian menstruation. *Acomys* is also studied as an evolutionary model of sympatric speciation (the evolution of a new species from an existing species while both continue to inhabit the same territory and exhibit overlapping ranges) in its native desert habitat in Evolution Canyon, Israel (Hadid et al. 2014; Wang et al. 2022).

There is a growing, but still small, number of research colonies of *Acomys* species distributed throughout the world and in these colonies *A. cahirinus* (Egyptian spiny mouse) is most often found. As adults, these animals are about twice the size of laboratory mice of the C57BL/6 J strain (*MusB6*), the litters are smaller (~ 2–4 pups/litter), and gestation times longer (~ 40–50 days) than *MusB6*. Newborn spiny mice are covered with hair, eyes are open, ears unfurled, and within 2 or 3 days after birth are moving around actively exploring their environment (Keller and Burns 1989; Regula 2012; Haughton et al. 2016; Pinheiro et al. 2018). By the age of 2 months, spiny mice are sexually mature (Regula 2012; Haughton et al. 2016). Spiny mice develop stiff and sharp-tipped hairs covering their dorsum that are modified awl hairs and are thought to serve a protective function in deterring predators (Montandon et al. 2014). Adults are characterized by large ears, a narrow snout, long whiskers, and large protruding black eyes (Fig. 1). At least three species of *Acomys* have been reported to develop skin that tears easily at low tensile forces (Seifert et al. 2012; Harn et al. 2021). While the underlying changes in embryonic skin morphogenesis that produce anatomically “shedtable” skin in adults of these species are not known, it is likely that acquisition of this developmental phenotype was a critical step in the evolution of *Acomys* species as it allowed escape from predators via autotomy in their native habitats.

Scarless wound healing in an adult mammal

Vertebrate species that are capable of regeneration as adults will heal acute tissue injuries without fibrotic scars (Brookes and Kumar 2008; Chen and Poss 2017; Stockdale et al. 2018; Maden and Varholick 2020). The majority of those vertebrate species are not mammals and do not have a fully committed terrestrial life style. Thus, the bulk of the regeneration literature comes from studies that use axolotl (a regeneration-competent species of salamander), newts & other urodeles, *Xenopus*, and zebrafish. Neonatal mouse myocardium is capable of limited regeneration potential after ventricular apex resection, although this potential is restricted to the first few postnatal days (Porrello et al. 2011). As mice are altricial at birth (born in an underdeveloped state requiring care and feeding by parents), neonatal heart regeneration may be related to the general scarless wound healing phenotype observed in mammalian fetal life (Clancy et al. 2001; Colwell et al. 2003).

Consistent with this, the same ventricular apex injuries heal with fibrotic scar by 7 days after birth and thereafter. Mouse digit tip regenerates after amputation in adults (Rinkevich et al. 2011; Lehoczký et al. 2011). Yet, it is only the extreme tip of the digit that regenerates limiting the overall relevance of this model for other tissues and organs. Current evidence suggests that spiny mice can regenerate a wide variety of tissues and organs as adults (Gaire et al. 2021; Okamura et al. 2022) and argues that *Acomys* should be added to the list of regenerators as a rare representative of the class Mammalia.

Skin

Given the damaging effects of tissue fibrosis on organ function (Wynn and Ramalingam 2012; Friedman et al. 2013; Humphreys 2018), the evolution of pathways for wound healing that minimize fibrosis and promote organ regeneration would seem advantageous (Fig. 2). The report by Seifert et al. showing that spiny mice can shed their dorsal skin to escape the grasp of predators and then fully regenerate the lost skin without fibrotic scarring suggests that *Acomys* species may have followed that evolutionary path (Seifert et al. 2012). These authors reported that the mean tensile strength of normal adult spiny mouse skin is at least 20-fold weaker than that of Swiss-Webster mice (Seifert et al. 2012). Moreover, skin shedding in spiny mice occurs by tearing without evidence of a pre-existing fracture plane, unlike the tail autotomy seen in geckos and other lizards (Baban et al. 2022; Ghatak 2022). This indicates that the composition of the extracellular matrix (ECM) of spiny mouse skin will be found to be considerably different than that of Swiss-Webster mice, as suggested by the opposing abundance ratios of collagens type I and type III in the two species and in the less dense and more porous arrangement of collagen fibers in spiny mouse skin (Seifert et al. 2012). Moreover, dermal fibroblasts isolated from dorsal skin of newborn *Mus* (CD1) and cultured on increasingly stiff matrices upregulated smooth muscle α -actin (*Acta2*) expression as a function of matrix stiffness, while newborn *Acomys* dermal fibroblasts do not upregulate *Acta2* expression when cultured under the same conditions of matrix stiffness (Stewart et al. 2018).

Tissue injury is typically accompanied by a localized innate immune response to promote wound healing (Eming et al. 2017; Moretti et al. 2022). A comprehensive analysis of the kinetics of leukocyte infiltration during ear hole regeneration in spiny mice was reported by Simkin et al. (2017). After determining that no significant differences in numbers of circulating monocytes, neutrophils, lymphocytes, and eosinophils preexisted in uninjured *Acomys* compared to *Mus* skin, the authors examined a time course of changes in leukocyte content in ear pinnae after injury. Over a 15 days time course, similar changes in CD11b-positive cells (neutrophils & macrophages) were found in *Acomys* and *Mus* ear tissues. Employing a pan-macrophage marker, IBA-1, no significant differences were found in the numbers of IBA-1-positive cells between the two species over the 15 days time course (Simkin et al. 2017). However, when lucigenin chemiluminescence was used to detect reactive oxygen species (ROS) production, much higher ROS levels were found in *Acomys* tissues at 1 days compared to *Mus* and this difference was maintained up to 5 days after injury (Simkin et al. 2017). Similar observations of sustained ROS production were reported for regeneration of zebrafish heart (Han et al. 2014) and *Xenopus* tadpole tail (Love et al. 2013) (reviewed in Helston and Amaya, 2021).

Macrophages are reported to be required for tissue regeneration in axolotl (Godwin et al. 2013), zebrafish (Lai et al. 2017), and neonatal mouse heart (Aurora et al. 2014). In a series of experiments to deplete phagocytes (mostly macrophages) using clodronate liposomes, Simkin et al. reported delays in blastema formation, cartilage histolysis, reepithelization, and closure of ear hole injuries in spiny mice (Simkin et al. 2017). These data provide evidence for a common cross-species requirement for macrophage function that is conserved in spiny mice. Precisely what mediators and macrophage subtypes are involved and what role they play in the cellular and molecular events underlying regeneration in spiny mice remain to be determined.

Although multiple cell types contribute to fibrotic matrix production, including perivascular cells (Passman et al. 2008; Kramann et al. 2015; Wu et al. 2016; Lu et al. 2020), endothelial cells (Yoshimatsu and Watabe 2022), and circulating fibrocytes (Reinhardt and Breuer 2021), a dominant role is played by the resident injury-activated myofibroblast (MF) (Hinz et al. 2012; van Putten et al. 2016; Henderson et al. 2020). Injury-induced MFs restore structural integrity of injured tissue but limit regeneration in most mammalian models (Darby et al. 1990; Tomasek et al. 2002; Hubert et al. 2021). In the skin, most MFs are derived from dermal fibroblasts (DFs) (Plikus et al. 2021; Talbott et al. 2022). DFs are now known to represent a number of distinct fibroblast subtypes that differ in embryonic origin, subtype-specific transcriptomes, anatomical position within skin tissue, and potentials for regeneration or scarring (Driskell et al. 2013; Rinkevich et al. 2015; Abbasi et al. 2020; Plikus et al. 2021; Talbott et al. 2022). Rinkevich et al. reported that DFs in the mouse arise from two independent origins in development that can be distinguished by whether or not their progenitor cells expressed *engrailed-1* (EN1) (Rinkevich et al. 2015). The EN1-positive lineage produces DFs that are the source of most of the connective tissue in developing skin and as well as the fibrotic matrix that appears in adult skin after injury (Rinkevich et al. 2015; Mascharak et al. 2021). Moreover, tissue mechanics interact with lineage-specific DF properties to influence the balance between fibrosis and regeneration (Mascharak et al. 2021; Harn et al. 2021). Recent multi-omics and single cell ATAC-seq studies of developing and adult skin point to a significant degree of plasticity among fibroblast subtypes. Genetic pathways for DF plasticity may have provided the substrate for evolutionary selection or amplification of DF states that mediate skin regeneration in *Acomys* species (Seifert et al. 2012; Brant et al. 2016; Abbasi et al. 2020; Foster et al. 2021; Plikus et al. 2021; Thompson et al. 2022).

Mechanical forces and ECM-cytoskeleton-nuclear tension play major roles in morphogenesis and wound healing (Wells and Discher 2008; Humphrey et al. 2015; Cho et al. 2017). The Hippo-Yap signaling pathway has been linked to mechanosensing in a wide variety of tissues and organs (Dupont et al. 2011; Wu and Guan 2021). To address the role of Hippo-Yap signaling in scarless wound healing, Brewer et al. 2021 examined Yap expression, localization, and function in repair of ear hole injuries in spiny mice (Brewer et al. 2021). No differences were found between *Acomys* and *Mus* dermal fibroblasts in Hippo pathway components or pathway activation upon cell-cell contact in vitro, a well-characterized activator of Hippo signaling (Gumbiner and Kim 2014; Brewer et al., 2021). Blocking Yap-TEAD interactions in vivo with verteporfin converted tissue regeneration to mispatterned repair and fibrosis in spiny mouse skin (Fig. 2). At the same time, however, studies in

MusB6 skin showed that verteporfin treatment of excisional wounds in dorsal skin prevented fibrosis and promoted tissue regeneration (Mascharak et al. 2021). Brewer et al. suggested that there are several possible explanations for these opposing findings. One, is that the normal matrix stiffness of mouse skin is much greater than *Acomys* skin, as discussed above (Seifert et al. 2012). Matrix stiffness is a known activator of Yap-dependent fibrosis genes and thus inhibition of Yap signaling in the mouse would be predicted to be anti-fibrotic. Second, *Acomys* produce a regeneration blastema during the initial states of wound healing (Gawriluk et al. 2016) whereas *Mus* (C57BL/6, CD1) do not. Third, Hippo-Yap signaling is well-known to be highly context-dependent (Moya and Halder 2019). Therefore, in the course of evolution of scarless wound healing the genomic Yap-TEAD binding sites may have become “rewired” in *Acomys* and that a comparative analysis of Yap-TEAD ChIP-seq datasets in *Acomys* versus *Mus* dermal fibroblasts during wound healing would provide important clues for how to achieve scarless wound healing in other mammals. A similar “rewiring” of glycosylation biosynthetic pathways was proposed by Nogueira-Rodrigues et al., for regeneration of spinal cord injury in *Acomys* (Nogueira-Rodrigues et al. 2022).

Skeletal muscle

A single intramuscular injection of cardiotoxin leads to a rapid breakdown of skeletal muscle fibers that is then restored by activation, proliferation and differentiation of resident muscle stem cells, classically known as satellite cells (Garry et al. 2016). Prior to injury, *Acomys* and *MusB6* were found to express similar levels of ECM proteins including collagen types I, IV, XII, XVII, fibronectin, laminin, and tenascin-C by immunofluorescence in tibialis anterior (TA) muscle (Maden et al. 2018). Collagen VI, however, was present in greater abundance in *Acomys* TA compared to *Mus* TA. Physiological stiffness of freshly isolated TA muscle was about twofold greater in *Mus* than in *Acomys*. Since cell migration, proliferation, and fibrosis are regulated by ECM stiffness, this difference in baseline muscle stiffness is noteworthy (Discher et al. 2005; Smith et al. 2017). Following a single injection of cardiotoxin, both spiny mice and C57BL/6 mice completely regenerated damaged TA muscle within 14 days with spiny mice regenerating muscle damage more rapidly than *Mus* (Maden et al. 2018). Regeneration occurred with similar changes in gene expression for myogenic transcription factors (*myf5*, *myoD1*, *myoG*) and TGFβ1, with higher levels of expression in *Mus*. Likewise, inflammation markers (including NF-κB) and ECM markers (collagen types I, III, & XII) were also increased in both species with much higher peak levels of expression in *Mus*. However, while cells expressing the macrophage M2 activation marker CD206 were present in equal numbers in both species at 4 days after cardiotoxin injury, cells expressing the M1 activation markers F4/80 or CD86 were present in abundance in damaged *Mus* muscle while they were undetectable in injured *Acomys* muscle (Maden et al. 2018).

Although repair of a single cardiotoxin injury was complete by 14 days in both species, striking differences between *Mus* and *Acomys* were observed after a series of five cardiotoxin injections each spaced 3 weeks apart. In this case, *Mus* muscle tissue regenerated very poorly after the fifth cardiotoxin injection with large numbers of adipocytes appearing among the few new muscle fibers that were formed (Maden et al. 2018). In contrast, skeletal muscle regeneration in spiny mice was robust with few, if any, adipocytes

appearing in the regenerated muscle tissue. Thus, in addition to the previously described regeneration of panniculus carnosus muscle in skin (Seifert et al. 2012; Brant et al. 2016; Gawriluk et al. 2016; Matias Santos et al. 2016; Brewer et al. 2021), spiny mice are capable of complete regeneration of TA skeletal muscle even after repeated, closely-spaced necrotic injuries in vivo (Maden et al. 2018).

Kidney

Clinically significant end-stage kidney disease and loss of organ function is strongly linked to progressive tissue fibrosis (Duffield 2014; Hill et al. 2016; Humphreys 2018). Fibrosis is a reliable predictor of renal functional decline (Coresh et al. 2007), and chronic kidney disease (CKD) is among the strongest risk factors identified for cardiovascular disease (Go et al. 2004). There are few treatment options currently available for patients with end-stage kidney disease who often require long-term dialysis or kidney transplantation for survival (Chaudhry et al. 2022).

To test whether scarless wound healing observed in spiny mouse skin extended to wound healing in vital internal organs, such as the kidney, two aggressive models of acute kidney injury were employed (Okamura et al. 2021). Unilateral ureter obstruction (UUO) is a surgical procedure resulting in complete ligation of the ureter. This procedure leads to renal tubular expansion with activation of resident tubular epithelial cells. Activated epithelial cells signal to interstitial fibroblasts and perivascular cells, leading to rapid initiation and expansion of interstitial fibrosis (Kramann et al. 2015; Humphreys 2018) (Fig. 3). This sequence of events results in rapid loss of organ function and eventual organ failure (Moriyama et al. 1998; Li et al. 2013). A second form of kidney injury that is more commonly encountered clinically is an ischemia–reperfusion injury (IRI) produced by clamping the vascular pedicle for a given amount of time followed by release of the clamp for kidney reperfusion (Singbartl et al. 2000; Pennathur et al. 2015; Liu et al. 2017). IRI also produces rapidly-developing interstitial fibrosis in mice leading to loss of kidney function and organ failure (Okamura et al. 2021). Yet spiny mice, while exhibiting similar acute kidney damage as either inbred C57BL/6 or outbred CD1 mice, fail to progress to interstitial kidney fibrosis with no significant increases in total collagen content or Sirius Red-positive areas on kidney sections. By contrast, *Mus* kidneys exhibited nearly tenfold increases in total collagen content with corresponding increases in Sirius Red-positive interstitial tissues. Remarkably, injured spiny mouse kidneys completely regenerated normal renal function by 2 weeks after either type of injury (Okamura et al. 2021) (Fig. 3). Regeneration of kidney function was accompanied by rapid clearance of tubular casts, resolution of inflammation, and absence of fibrotic tissue formation. Transcriptomic analysis 2 days after obstructive injury suggested that the spiny mouse genome is poised to initiate a regenerative response at the time of the initial injury event (Okamura et al. 2021). For example, RNA seq analysis at 5 days after UUO injury revealed re-expression of a nephrogenic gene expression profile with rapid expansion of patches of cadherin-6-positive tubular epithelial cells in *Acomys*, but not in *Mus*, that correlated with regeneration of kidney function in spiny mice. Unlike ear hole injuries in skin, a regeneration blastema was not found in spiny mouse kidneys.

Heart

The response of spiny mouse hearts to myocardial ischemia (MI) following coronary ligation has been examined in three recently published studies (Qi et al. 2021; Koopmans et al. 2021; Peng et al. 2021). The results of the three studies are highly concordant with the common findings being: (1) spiny mouse hearts exhibit increased resistance to ischemic injury compared to *Mus*, (2) a robust angiogenic response after MI is observed in the infarct zone that persists over time in spiny mice, (3) a minimal infarct “scar” appears in spiny mouse hearts that is more porous and compliant than fibrotic scars in *Mus* hearts, and (4) as assessed by ultrasound imaging, spiny mouse hearts exhibit an *improvement* in cardiac function over time after MI while similarly injured *Mus* hearts exhibit a progressive *decline* in cardiac function over the same time period.

For example, Qi et al. showed that permanent ligation of the left main anterior descending coronary artery in adult spiny mice produced an acute drop in the ejection fraction by 1 day after MI to levels about 25% of uninjured controls (Qi et al. 2021). This reduction in ejection fraction was very similar to that seen in *Mus* hearts after MI for the same time period consistent with equivalent acute cardiac injuries sustained by both species. Over the next 4 weeks, spiny mice restored ejection fraction values to about 65% of uninjured controls while *Mus* hearts exhibited no improvement. By 13 weeks after MI, *Mus* hearts showed a 37% decrease in wall thickness of the left ventricular free wall while spiny mouse hearts exhibited no decrease in wall thickness. Qi et al. also reported that spiny mouse hearts had a smaller infarct zone with greatly reduced fibrotic scar area than did *Mus* hearts over the same time period (Qi et al. 2021). Using micro-CT analysis, *Acomys* hearts were found to have formed a much more substantial vascular network within the infarct zone with conduit-like arteries distributed throughout the fibrotic “scar” tissue indicative of high rates of blood flow. These vascular features in spiny mouse hearts departed sharply from the sparse distribution of small diameter vessels in the infarct zone of *Mus* hearts with large areas of scar tissue appearing avascular. Following daily injections of 5-bromo-2'-deoxyuridine (BrdU) starting at day 3 and continuing until day 14 after MI, Qi et al. reported that up to 20% of the BrdU-labeled cells in infarcted spiny mouse hearts were also labeled by sarcomeric α -actin and provisionally identified as proliferating cardiac myocytes (Qi et al. 2021).

Koopmans et al. similarly reported that the infarct “scar” tissue in spiny mouse hearts is substantially different from the familiar dense collagen-rich fibrotic scar that forms in similarly injured *Mus* hearts (Koopmans et al. 2021; Fu et al. 2018; Yokota et al. 2020). The collagen fibers in *Acomys* hearts are arranged in a highly porous “basket weave” pattern that has higher compliance and allows for greater ventricular conduction with minimal resistance to contraction than *Mus* scar tissue, features that are consistent with recovery of ejection fraction in spiny mouse hearts (Koopmans et al. 2021). By contrast, collagen fibers in *Mus* infarct scars are densely packed in parallel arrays that resist compression during ventricular contraction and lead to progressive loss of cardiac function as the scar tissue expands and matures over time (Fu et al. 2018; Yokota et al. 2020; Burke et al. 2021). Since neonatal mice are known to possess cardiac regeneration potential (Porrello et al. 2011), Koopmans et al. compared cardiac transcriptomes between adult spiny mouse hearts and

neonatal *Acomys* and *Mus* hearts (Koopmans et al. 2021). Principal component analysis produced the intriguing result that adult spiny mouse hearts exhibit very little difference in overall gene expression when compared to neonatal spiny mouse cardiac transcriptomes suggesting that adult *Acomys* cardiac myocytes exist in an immature state. These results contrast with the comparison of neonatal versus adult *Mus* hearts where many differences in gene expression were found, corresponding to the maturation steps known to take place in the transition from newborn to adult mouse cardiac physiology via SRF-MRTF signaling (Guo et al. 2021; Miano et al. 2007). Taken together, these findings suggest that adult spiny mouse hearts maintain a neonatal pattern of gene expression with a minimal signature of postnatal maturation events.

Zebrafish hearts are known to regenerate apex resection injuries, and cardiac myocyte proliferation is a prominent component of the regenerative response (Kikuchi et al. 2010). Unlike adult mammalian hearts, zebrafish hearts maintain a predominant mononucleate cardiomyocyte population that serve as progenitors for the cells that proliferate after injury (Kikuchi and Poss 2012). Peng et al. reported a higher frequency of mononucleated cardiomyocytes in normal adult *Acomys* hearts (26%) compared to adult *Mus* hearts (6%) (Peng et al. 2021). Like the studies cited above, Peng et al. also found spiny mouse hearts are resistant to ischemic injury, develop increased vascular density in the infarct zone with proliferating endothelial cells and Acta2-positive mural cells, and produce a more porous and compliant “scar” tissue (Peng et al. 2021). Each of these features would contribute to an adult spiny mouse heart that is better able to preserve and restore cardiac function after MI.

Spinal cord

Injury to the spinal cord initiates a sequence of inflammatory responses involving microglial activation, influx of neutrophils and monocytes that migrate to the injury site, and formation of a glial/fibrotic scar that acts as a physical and chemical barrier to axon regeneration (Dias et al. 2021). Spiny mice exposed to spinal cord crush injury at the C3 level had reduced production of proinflammatory cytokines, increased expression of neurogenesis genes, absence of glial scar formation, rapid restoration of ability to void the bladder, restoration of weight-bearing and locomotion in hind limbs, and reduced TGF β 1 signaling compared to *Mus* with a similar injury (Streeter et al. 2020). The authors concluded that a blunted immune response and failure to produce fibrotic tissue may be characteristic, not only of spinal cord repair, but of wound healing in general for other spiny mouse tissues and organs (Streeter et al. 2020).

In a study of complete spinal cord transection at the T8 level, Nogueira-Rodrigues et al. reported that spiny mice form a “bridging matrix” across the lesion site whereas *Mus* produce a fibrotic scar (Nogueira-Rodrigues et al. 2022). While spiny mice develop numerous β III-tubulin-positive axons, some fully myelinated, spanning the bridging matrix by 8 weeks after transection, *Mus* formed few, if any, such axons. Compound action potentials could be measured across the lesion site in 5 out of 10 injured spiny mice by 8 weeks while no signal conduction was detected in *Mus* at 8 weeks. As in skin, kidney, heart, and muscle, there were reduced levels of collagen fiber content by Masson’s Trichrome staining in the bridging matrix produced at the injury site in spiny mice while abundant

densely-packed collagen fibers were found in *Mus*. These conclusions were supported by collagen type 1 immunostaining data spanning the injury site in the two species. RNA-seq analysis pointed to strong differences in acetylglucosaminyltransferase enzyme levels which included upregulation of keratan sulfate proteoglycan (KSPG) biosynthetic enzymes and downregulation of heparan sulfate proteoglycan (HSPG) biosynthetic enzymes in injured spiny mice spinal cords compared to *Mus*. In particular, β -1,3-N-acetylgalactosaminyltransferase 7 (β 3gnt7) and carbohydrate sulfotransferase-15 (Chst15) were strongly increased in spiny mice compared to *Mus* (Nogueira-Rodrigues et al. 2022). β 3gnt7 is a critical enzyme for KSPG production while Chst15 is required for chondroitin sulfate proteoglycan (CSPG) synthesis. When Chinese Hamster Ovary (CHO) cells overexpressing β 3gnt7 were used as a substrate and *Mus* dorsal root ganglia or cortical neurons were plated on top of the CHO cells in vitro then greatly increased neurite and axon outgrowth was observed compared to vector only controls.

Taken together, these two studies show that spiny mice, unlike any other mammalian species tested, exhibit substantial regeneration of spinal cord injury and remarkable recovery of hind limb function after spinal cord transection (Streeter et al. 2020; Nogueira-Rodrigues et al. 2022; Wehner and Becker 2022). These data provide hope for the ~ 300,000 patients living with spinal cord injuries in the United States alone (Jain et al. 2015) that understanding spinal cord regeneration in *Acomys* may lead to more effective therapies in the near future.

Hibernating mammals

There is a curious parallel between wound healing in spiny mice and that reported for hibernating mammals. For example, Iaizzo et al. observed that hibernating black bears heal full thickness skin wounds with little or no infection, de novo formation of hair follicles, and an absence of fibrotic scars (Iaizzo et al. 2012). This is similar to the observations made of skeletal muscle injuries in hibernating 13-lined ground squirrels where healing occurs without fibrotic scars during hibernation (Andres-Mateos et al. 2012). The wound repair process is then completed with hair formation once the animals emerge naturally from hibernation (Andres-Mateos et al. 2012). These two examples come from cases of winter hibernation where core body temperatures are reduced compared to the awake state.

Yet a similar phenomenon also occurs in response to physiological stresses encountered in the native hot and dry environments that spiny mice normally inhabit. For example, a state of decreased metabolic activity, reduced heart and respiratory rate, and core body temperature called torpor is acquired by golden spiny mice (*Acomys russatus*) exposed to elevated temperatures and extended periods of food and water shortage commonly encountered in their native desert habitat (Grimpo et al. 2013; Mohr et al. 2020). In this case, a 50% food reduction led to a decrease in daily energy expenditure that was attributed to the metabolic effect of daily torpor (Grimpo et al. 2013). This physiologic state thus lowers metabolic activity in proportion to reduced food and water availability. Torpor therefore allows golden spiny mice to survive in their harsh native desert environment (Grimpo et al. 2014). Although these authors did not monitor wound healing during torpor, their findings raise the intriguing possibility that the unusual wound healing properties of adult spiny mice may have been achieved by gaining transcriptional access to scarless wound healing

pathways without a requirement for physiological torpor or a hibernation-like metabolic state. Wound healing in hibernating mammals may therefore provide additional support for the idea that regeneration is a more broadly distributed property of mammalian genomes than is currently thought.

The spiny mouse genome

Wang et al. recently published a fully sequenced, assembled, and annotated *Acomys cahirinus* reference genome (Wang et al. 2022). These authors reported the spiny mouse genome has a total size of 2.27 Gb with 21,879 coding genes and 19 pseudochromosomes. They identified 586 distinct miRNAs, 1327 small nuclear RNAs, and ~ 878 Mb (39%) repetitive sequences (Wang et al. 2022). A phylogenetic tree analysis based on 6285 single-copy orthologous genes showed *A. cahirinus* was most closely related to *A. dimidiatus* followed by *A. kempi*, *A. percivali* and then *A. russatus*. Comparing the spiny mouse genome (*A. cahirinus*) to that of *Mus musculus*, 200 genes under positive selection in spiny mice were identified that were enriched in Gene Ontology (GO) terms related to embryonic morphogenesis and brain development (Wang et al. 2022). This intriguing finding is consistent with the response to injury studies reviewed above and is particularly relevant to the findings reported by Okamura et al. who identified re-expression of a set of nephrogenic progenitor genes in adult spiny mouse kidney 5 days after unilateral ureter obstruction injury (Okamura et al. 2021).

Sympatric speciation, first proposed by Darwin, is the evolution of a new species from an existing species while both continue to inhabit the same territory and exhibit overlapping ranges (Darwin 1859). Two populations of spiny mice appear to be undergoing sympatric speciation on the opposing slopes of Evolution Canyon, Israel (Hadid et al. 2014). These populations are under divergent selection as the European slope (ES) is exposed to morning sun, is cooler, and more densely forested whereas the opposite slope, the African slope (AS), is exposed to hot afternoon sun, is dryer, and is only sparsely forested (Li et al. 2016). To address the potential role of epigenetic regulation in relation to sympatric speciation in Evolution Canyon, Wang et al. carried out whole genome bisulfite sequencing and acquired DNA methylation patterns from their long-read DNA sequences (Oxford Nanopore Technologies, UK) of spiny mice from ES and AS. They report that an analysis of 28,636 differentially methylated regions revealed four functional enrichment categories related to sympatric speciation including DNA damage repair (Matsuoka et al. 2007), kidney system development, lipid storage and usage, and circadian rhythm (Wang et al. 2022).

Insights for mammalian organ regeneration

Progressive tissue fibrosis has proven to be an elusive therapeutic target. Ongoing mechanistic investigation continues to identify new methods and molecular players to unravel the pathogenesis of fibrosis that may provide effective targets for intervention (Doherty et al. 2010; Travers et al. 2022). Yet, a viable complementary direction is to look to nature for variant wound healing species that might be informative. For mammals, evolution of the genus *Acomys* has provided a potentially powerful resource for clues as to how to redirect mammalian wound healing from chronic inflammation and fibrosis to scarless

wound repair and complex tissue regeneration. There are countless examples in nature where evolution by natural selection has produced innovative solutions to complex and difficult biological problems. For spiny mice, because the “solution” has evolved within the full complexity of a mammalian genome and mammalian physiology, we would anticipate fewer obstacles to clinical translation than many of the current approaches have encountered.

Summary and future directions

The studies reviewed above suggest that the spiny mouse (genus: *Acomys*) is an emerging model organism with exciting potential to identify mechanisms of complex tissue and organ regeneration that operate in the context of adult mammalian physiology (Fig. 4). Scarless wound healing with regeneration of mammalian organ function is a long-sought goal of the field of regenerative medicine. The remarkable wound healing properties that have evolved in the genus *Acomys* promise unique insights with potential clinical applications for mammalian tissue regeneration. Looking ahead, future studies will address (1) identification of the epigenetic controls for retention versus loss of fetal scarless wound healing phenotypes in *Acomys* versus *Mus*; (2) a more complete characterization of the composition and physical properties of the ECM in *Acomys* tissues and organs; (3) a more detailed investigation of the differences in immune responses to injury in *Acomys* versus *Mus*; and (4) an identification of cis-regulatory sequence changes that alter the levels, timing, or cell type specificity of the expression of key genes determining tissue responses to injury in spiny mice. These, and other studies, will provide additional key insights into the evolution of scarless wound healing in an adult mammal.

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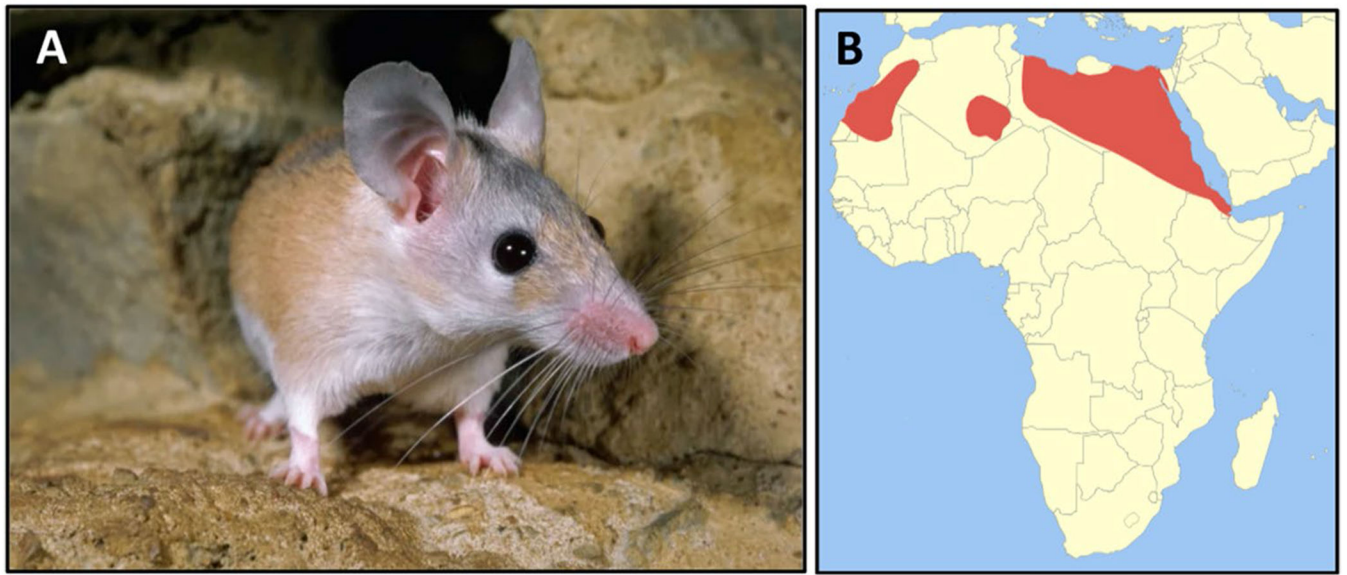
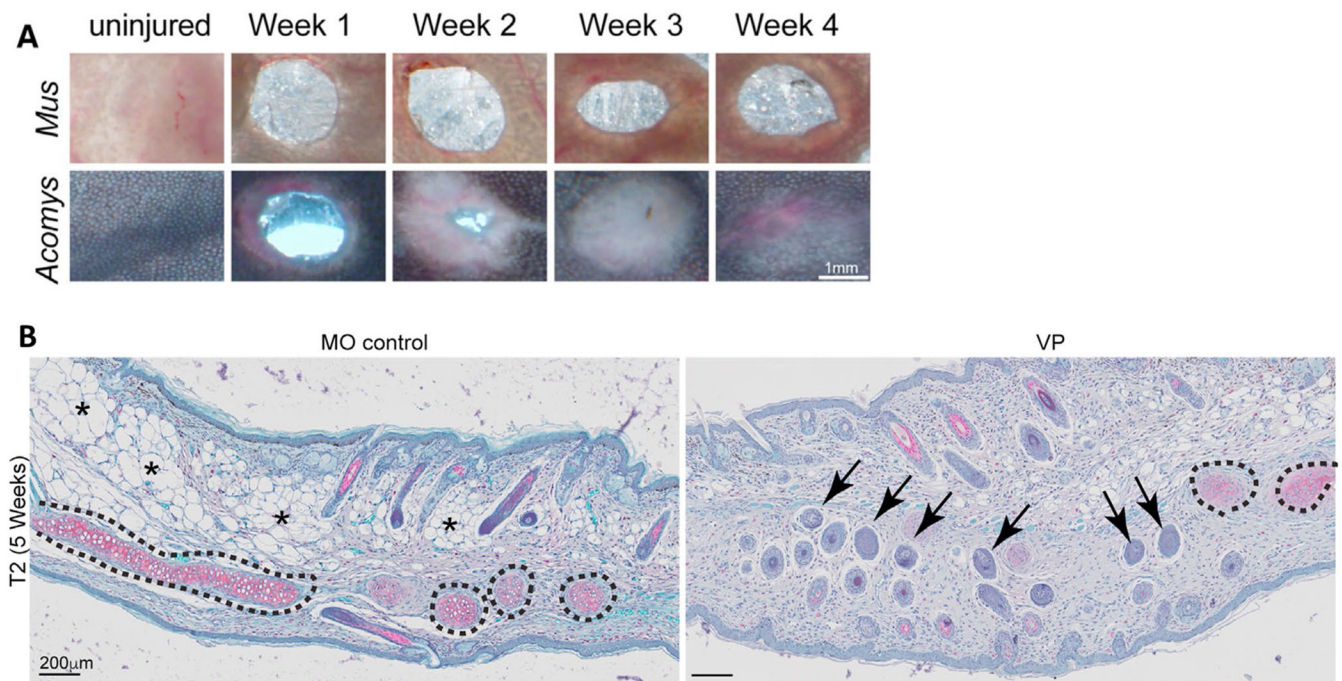


Fig. 1.
Acomys cahirinus—spiny mouse. **A** Adult spiny mice have narrow snouts, long whiskers, large ears, protruding black eyes, and spiny hairs along the dorsum. **B** These animals are adapted to hot arid climates and are found throughout northern Africa (red) and the Middle East (image licensed under the Creative Commons Attribution-Share Alike 3.0 Unported license-Wikimedia Commons)

**Fig. 2.**

Regeneration of ear hole injury in adult spiny mice. **A** Time course of ear hole closure and repigmentation in *Mus* (CD-1, top) and *Acomys* (spiny mouse, bottom). Note complete regeneration of ear hole injury in *Acomys* together with pigment and hair formation. **B** Histology (Safranin O stain). Left panel shows regeneration of normal ear architecture in vehicle-treated (mineral oil, MO) spiny mice with new cartilage (black dashed lines), adipocytes (asterisks), and hair follicles. (right panel) After treatment of *Acomys* with verteporfin (VP), an inhibitor of Yap-TEAD binding interactions (Brewer et al. 2021) between 3 and 5 weeks after ear punch injury, the resulting regenerated tissue is mispatterned and fibrotic. Significant reductions of new cartilage (dashed black lines) and adipocytes together with numerous ectopically located hair follicles appearing in the ventral region of de novo ear tissue at 5 weeks after injury (Brewer et al., 2021). Reprinted with permission from *Developmental Cell*

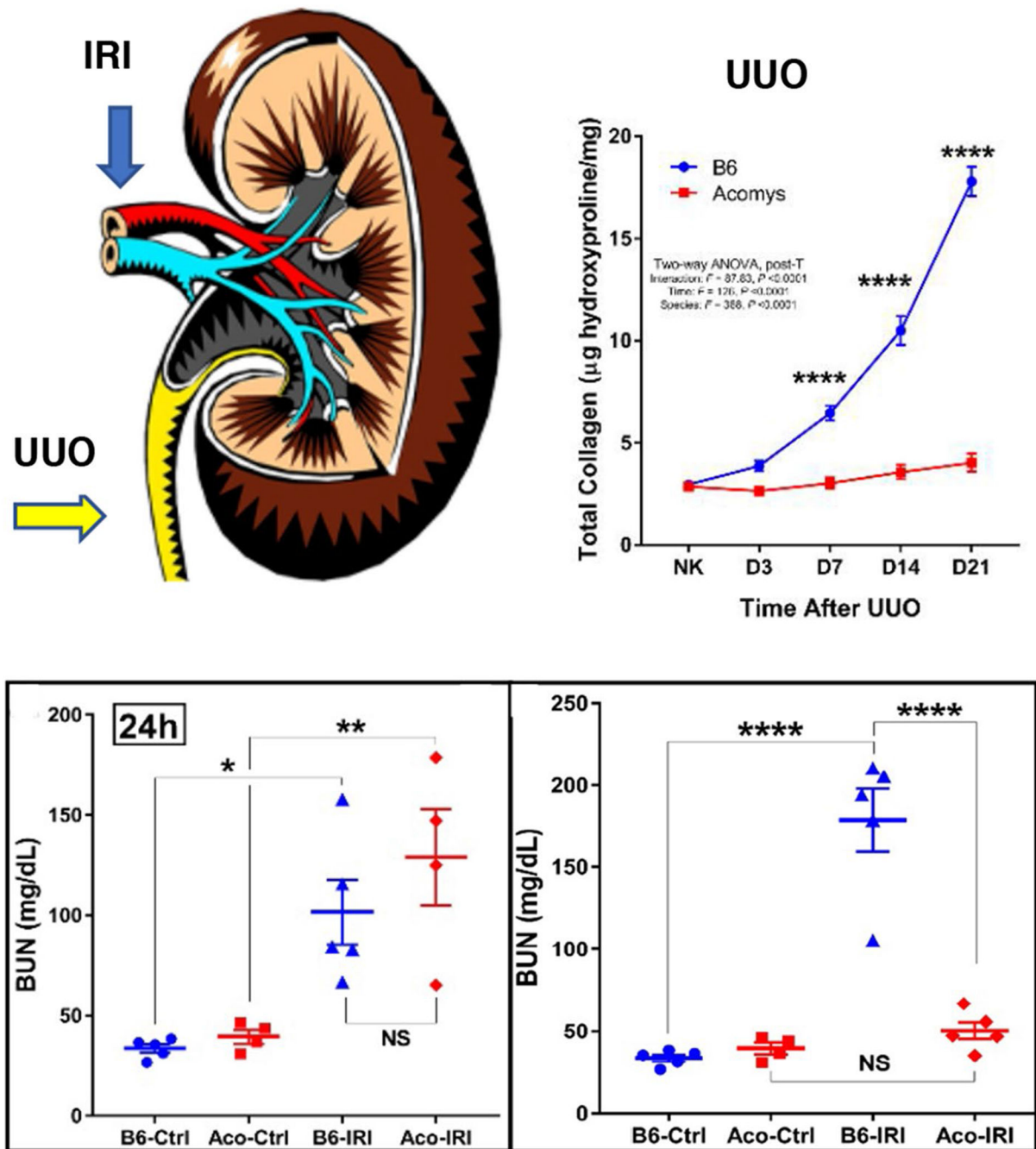


Fig. 3.

Regeneration of kidney function after acute renal injury in *Acomys*. (*top left*) Kidneys were injured either by permanent obstruction of the ureter (UUO, yellow arrow) or clamping of the renal artery for 40 min followed by release of the clamp allowing reperfusion of the ischemic kidney (IRI, blue arrow). (*top right*) Absence of fibrosis in obstructed kidneys by total collagen content measured as micrograms hydroxyproline per mg wet kidney weight in *Acomys* (red). By contrast, UUO produces a robust fibrotic response in *Mus*^{B6} (blue) kidneys. (*bottom left*) Kidney function measured by blood urea nitrogen (BUN) levels 24 h

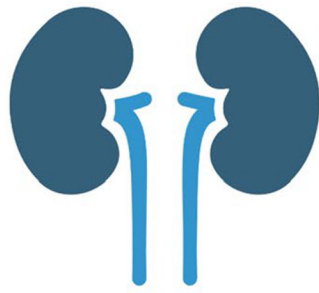
after IRI injury. Note that both *Acomys* (Aco, red) and *Mus^{B6}* (blue) exhibit similar levels of elevated BUN indicative of acute loss of kidney function. (*bottom right*) BUN levels at 2 weeks after IRI injury. Note further declines in kidney function in *Mus^{B6}* (blue) progressing to organ failure. By contrast, BUN levels in *Acomys* (red) have returned to normal baseline values indicative of complete regeneration of organ function. *Ns* not significant, **p* < 0.05, ***p* < 0.01, ****p* < 0.001, *****p* < 0.0001. (Okamura et al. 2021). Reprinted with permission from *iScience*. Figure created in BioRender

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Acomys cahirinus - spiny mouse

resists oxidative injury
minimal or no fibrosis
reduced inflammation
robust angiogenesis
immature phenotype of adult cells
genome poised for regeneration

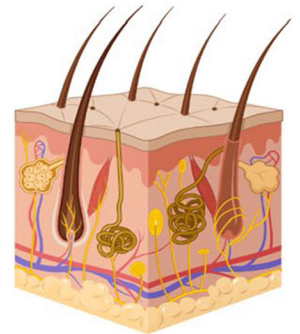
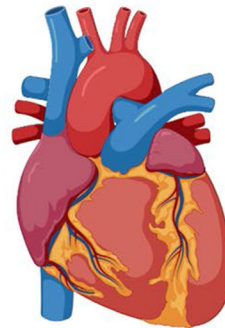
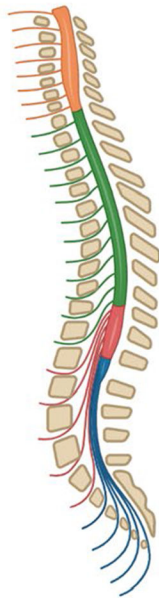


Fig. 4.

Acomys cahirinus (spiny mouse) (center top panel) is an emerging model organism to investigate and clinically translate molecular mechanisms of mammalian organ regeneration. Comparative studies have been reported of restored organ function after acute injury (from left to right) in kidney (Okamura et al. 2021); spinal cord (Nogueira-Rodrigues et al. 2022; Streeter et al. 2020), heart (Koopmans et al. 2021; Peng et al. 2021; Qi et al. 2021), skin (Brant et al. 2016; Brewer et al. 2021; Gawriluk et al. 2016; Harn et al. 2021; Matias et al. 2016; Jiang 2019; Seifert et al. 2012) and skeletal muscle (Maden et al. 2018). Some of the common findings across all studies of spiny mouse wound healing are indicated in the center boxed text. Figure created in BioRender